Predicting the Final Sale Price of a House in Irvine Using CRISP-DM

Introduction

A real estate company wants to predict the final sale price of a house in Irvine. The goal is to help agents and sellers make better pricing decisions before listing a property.

1. Business Understanding

The business goal is to predict the final sale price of a house. The project should provide accurate predictions to support pricing decisions.

Risks: Changing market conditions and unclear business requirements.

2. Data Understanding

I would collect historical home sales data, including location, square footage, bedrooms, bathrooms, lot size, house age, HOA fees, and final sale price. I would also check for missing values and data quality.

Risks: Missing or inaccurate data.

3. Data Preparation

I would clean the data, remove duplicates, handle missing values, and create new features such as house age and price per square foot. The data would then be divided into training and test sets.

Risks: Data leakage and incorrect data cleaning.

4. Modeling

I would build a Linear Regression model to predict the final sale price based on the property features.

Risks: Overfitting and selecting inappropriate variables.

5. Evaluation

I would test the model using five recently sold homes in the same Irvine neighborhood. Their property features would be entered into the model, and the predicted prices would be compared with the actual sale prices to evaluate the model's accuracy.

Risks: The five houses may not represent the entire market, and recent market changes may affect the prediction accuracy.

6. Deployment

The final model could be added to a real estate system. Agents could enter a home's information and receive a predicted final sale price. The model should be updated regularly as the housing market changes.

Risks: Outdated data and changing market conditions.

Conclusion

By following the CRISP-DM process, the company can build a reliable model to predict the final sale price of houses in Irvine and support better real estate decisions.